HEALTHCARE IoT NETWORK SECURITY & INTRUSION DETECTION SYSTEM

University of Cincinnati | EECE Senior Capstone | Spring 2026

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01 | PROJECT OVERVIEW

Healthcare networks face unprecedented cybersecurity challenges, with 82% of organizations experiencing IoT-related security incidents and ransomware attacks disrupting critical care operations. This senior capstone project implements a comprehensive healthcare IoT network simulation with integrated security monitoring, demonstrating both vulnerable and hardened architectures through physical demonstrations.

Our system consists of a fully operational patient vital monitoring pipeline running on physical hardware: five Raspberry Pi devices simulate cardiac patients transmitting real-time heart rate data via MQTT to a centralized data infrastructure. The pipeline processes 100 readings per second through Node-RED, stores data in PostgreSQL with 7-day retention windows, converts to HL7 FHIR standard via Mirth Connect, and visualizes through Grafana dashboards with Keycloak single sign-on authentication.

The project culminates in a live attack demonstration comparing vulnerable vs. secure IoT architectures. Using two identical pneumatic control systems, we show how sensor spoofing attacks can manipulate physical systems (causing catastrophic failure on the vulnerable system) while multi-layer validation successfully defends the hardened system. Suricata IDS, Wazuh SIEM, and Elasticsearch provide real-time attack detection and forensic analysis.

THE CHALLENGE: Healthcare Cybersecurity Crisis

INDUSTRY STATISTICS

- 25% of all ransomware attacks target healthcare (highest any industry)
- $10.9M average breach cost — 2.5× higher than other sectors
- 82% of healthcare orgs experienced IoT incidents (2023)
- Medical device recalls due to cyber vulns increased 138% (2022–23)

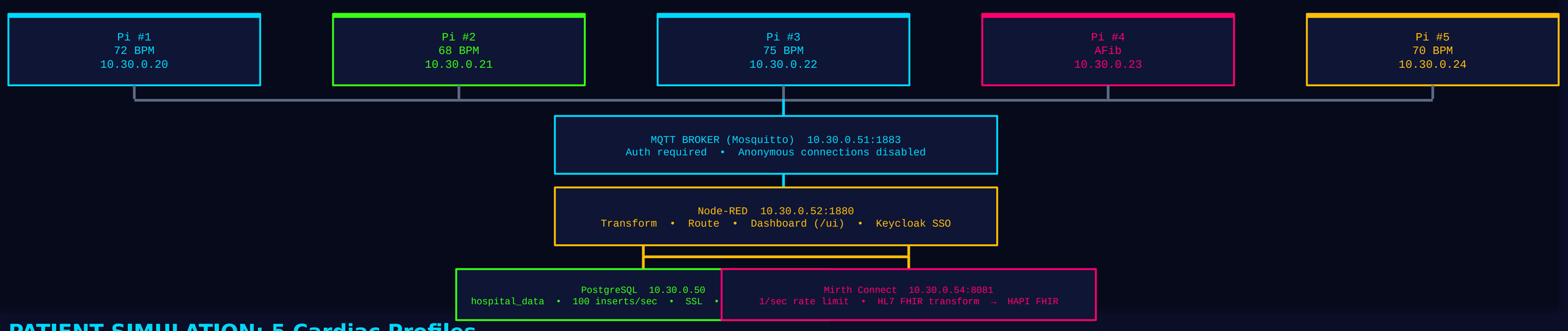
REAL-WORLD CONSEQUENCES

- 2020: UHS ransomware shut down 400 facilities, rerouted ambulances, delayed surgeries
- 2021: FDA recalled 465,000 pacemakers — network vulns for unauthorized firmware updates
- 2022: Insulin pump vulnerability allowed remote dosing manipulation via unauthenticated BT
- 2023: IoT botnet infections shut down hospital surgical units and patient monitoring systems

SOLUTION: Multi-Layer Defense Architecture

1. Network Segmentation:	VLAN30 (secure) / VLAN2 (attack lab) with UniFi UDM-SE firewall-enforced access controls
2. Real-Time Monitoring:	Suricata IDS with port mirroring on Switch Port 15; Filebeat ships to Elasticsearch :9200
3. SIEM:	Wazuh SIEM with 14 agents — correlates events, triggers anomaly alerts across all hosts
4. Identity & Access Mgmt:	Keycloak SSO (10.30.0.57) — centralized auth for Grafana, Node-RED, HAPI FHIR, Wazuh, Vault
5. Secrets Management:	HashiCorp Vault (10.30.0.58) — 3-of-5 unseal keys, OIDC via Keycloak, audit logging
6. Data Encryption:	PostgreSQL SSL/TLS, MQTT auth required, all sensitive data encrypted at rest and in transit
7. Physical Validation:	Sensor data cross-validated against ground truth — demonstrated via balloon attack scenario

HOSPITAL HOMELAB: Complete Data Pipeline



PATIENT SIMULATION: 5 Cardiac Profiles

#	Profile	Rate / Pattern	IP Address	Script	MQTT Topic
Pi #1	Stable Patient	72 BPM ± 2.5 HRV	10.30.0.20	HeartRate1Monitor.py	heartrate/pi-1/reading
Pi #2	Variable Patient	68 BPM ± 3.5 HRV	10.30.0.21	HeartRate2Monitor.py	heartrate/pi-2/reading
Pi #3	Highly Stable Patient	75 BPM ± 2.0 HRV	10.30.0.22	HeartRate3Monitor.py	heartrate/pi-3/reading
Pi #4	Atrial Fibrillation Sim	70–150 BPM irregular	10.30.0.23	HeartRate4Monitor.py	heartrate/pi-4/reading
Pi #5	Rare Anomaly Patient	70 BPM, ~1 anomaly / 5.5hr	10.30.0.24	HeartRate5Monitor.py	heartrate/pi-5/reading

INFRASTRUCTURE COMPONENTS

PostgreSQL	10.30.0.50	Ubuntu 24.04, PostgreSQL 16.10, 1TB storage, SSL enabled	**HAPI FHIR**	10.30.0.56	Docker, port 8080, OAuth2 Proxy :4180
MQTT Broker	10.30.0.51	Mosquitto — authenticated (mqtt_client), anonymous disabled	**Keycloak**	10.30.0.57	Port 8080, realm: hospital-homelab, SSO for all services
Node-RED	10.30.0.52	Dashboard :1880/ui, Keycloak SSO integration	**HashiCorp Vault**	10.30.0.58	Port 8200, 3-of-5 unseal keys, OIDC via Keycloak
Grafana	10.30.0.53	Port 3000, Keycloak SSO, UTC display (5hr ahead EST) — queries PostgreSQL 10.30.0.50 for dashboard data	**Elasticsearch**	10.30.0.60	Port 9200, TLS, receives Suricata logs via Filebeat
Mirth Connect	10.30.0.54	HTTPS :8443, HL7 FHIR channel :8081, rate limited 1/sec	**Suricata IDS**	192.168.50.191	Bare metal Ubuntu (VLAN2) — mirrors Switch Port 15 → Port 7, Filebeat → 10.30.0.60:9200
Wazuh SIEM	10.30.0.55	14 agents, Keycloak SSO, central event correlation			

SYSTEM PERFORMANCE

100/sec	**~53M**	**7 days**	**87 GB**	**1/sec**	**30 sec**
PostgreSQL insert rate	heartrate_data rows	active retention window	historical data archive	Mirth Connect FHIR rate limit	Grafana refresh interval

WHAT WE DEMONSTRATE

1. **Fully Operational Healthcare IoT Pipeline:** Build a realistic patient monitoring system with enterprise-grade infrastructure — production-ready architecture with high availability, data retention policies, and backup strategies.
2. **Defense in Depth Security Architecture:** Implement multiple overlapping security controls (network segmentation, IDS, SIEM, authentication, encryption, audit logging) following NIST cybersecurity framework.
3. **Compliance & Standards:** Demonstrate HIPAA-relevant controls (encryption, audit trails, access controls) and healthcare interoperability standards (HL7 FHIR, LOINC coding).
4. **Physical Attack Demonstration:** Prove cybersecurity controls work through tangible demonstration — vulnerable IoT system fails catastrophically while hardened system successfully defends (Posters 4–6).
5. **Real-Time Detection & Response:** Show complete incident detection pipeline: network capture (Suricata) → event correlation (Wazuh) → visualization (Grafana) with sub-second alert latency.
6. **Scalable, Maintainable Infrastructure:** All components containerized or virtualized for portability, configuration managed via infrastructure-as-code principles, services auto-restart on failure.